1. Perform a simple calculation

Relevance: ★★★★★ (Highest)

- The core of the problem is a **percentage calculation**
- Only basic arithmetic operations are involved
- No complex modelling or reasoning is required

👉 This is the **central mathematical activity** of the task.

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2. Manipulate numbers, graphical and statistical data and information

Relevance: ★★★★★

- The problem requires comparing two numerical values (78 and 110)
- Calculating the difference and expressing it as a proportion
- A typical example of **numerical manipulation and comparison**

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3. Use a multi-step procedure leading to a mathematical solution

Relevance: ★★★★☆

Although straightforward, the solution involves several steps:

1. Understanding the phrase "how many percent heavier"
2. Finding the difference between the two values
3. Using the original value as the denominator
4. Converting the result into a percentage

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4. Understand and utilise constructs based on definitions, rules and formal systems

Relevance: ★★★★☆

- Correct interpretation of the definition of **percentage increase**
- Recognising that the denominator should be the **first egg (78 g)**
- This is a common source of error if the definition is misunderstood

5. Implement a given strategy to determine a mathematical solution

Relevance: ★★★☆☆

- Applying a familiar strategy:

  > Percentage increase = increase ÷ original amount

- The strategy itself is standard and direct

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6. Use an understanding of the context to guide or expedite the mathematical solving process

Relevance: ★★★☆☆

- The phrase “the second egg is heavier than the first” clarifies the direction of comparison
- The biological context does not affect the calculation but **supports comprehension**

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7. Articulate a solution, showing and/or summarising intermediate mathematical results

Relevance: ★★★☆☆

- Even though it is a multiple-choice question, a complete solution involves showing intermediate steps
- Particularly relevant for assessing **students’ mathematical communication skills**

① Perform a simple calculation（进行简单计算）

相关性：★★★★★（最高）

- 本题核心是一个**百分比计算**
- 只涉及加减、除法和百分数
- 没有复杂建模或推理

👉 这是题目的**核心数学活动**

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② Manipulate numbers, graphical and statistical data and information（数值操作）

相关性：★★★★★

- 对两个给定数值（78、110）进行比较
- 计算差值并进行比例运算
- 属于典型的**数值处理与比较**

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③ Use a multi-step procedure leading to a mathematical solution（多步骤求解）

相关性：★★★★☆

虽然不复杂，但仍然是**多步过程**：

1. 理解“比……重百分之多少”
2. 计算差值
3. 用原始值作分母
4. 转换为百分数

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④ Understand and utilise constructs based on definitions, rules and formal systems（理解并运用数学定义和规则）

相关性：★★★★☆

- 需要正确理解**“A 比 B 多百分之多少”**的定义
- 明确**分母应是第一个蛋（78g）**
- 很多学生易在此出错（误用 110 作分母）

⑤ Implement a given strategy to determine a mathematical solution（执行已知策略）

相关性：★★★☆☆

- 使用常见策略：

  > 百分比增长 = 增量 ÷ 原始量

- 策略是熟悉且直接的

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⑥ Use an understanding of the context to guide the mathematical solving process（利用情境理解指导解题）

相关性：★★★☆☆

- “第二个蛋比第一个蛋重” → 明确比较方向
- 生物背景并不用于计算，但**帮助理解问题问法**

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⑦ Articulate a solution, showing intermediate mathematical results（表达解题过程）

相关性：★★★☆☆

- 虽然是选择题，但完整解答需呈现中间步骤
- 特别适合用于考查学生的**解题表达能力**